

Detecting and Mitigating Algorithmic Bias in Credit Models: A Tutorial Using Public Financial Data

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Abstract—This tutorial presents a practical walk-through for detecting and mitigating algorithmic bias in credit scoring models using public financial datasets. We demonstrate the application of fairness metrics and mitigation techniques via open-source toolkits such as AIF360 and Fairlearn, with a focus on responsible and equitable AI in financial decision-making.

Index Terms—Algorithmic Fairness, Credit Scoring, Bias Mitigation, Financial Data, Data-Centric AI, AIF360, Fairlearn

I. INTRODUCTION

The adoption of automated decision-making systems in financial services has expanded rapidly, particularly in credit scoring and loan approval workflows. Machine learning models are increasingly used to assess creditworthiness based on historical applicant data. While these systems offer efficiency and scalability, they also inherit risks embedded in the data on which they are trained. If sensitive or socially correlated features—such as gender, age, or nationality—are present, either directly or through proxies, the model may produce outcomes that disproportionately disadvantage certain groups.

This concern has brought algorithmic fairness to the forefront of AI governance and risk management. In credit decision contexts, unfair treatment can lead not only to reputational and ethical challenges, but also to regulatory non-compliance. Traditional approaches to fairness in machine learning often focus on removing protected attributes from the dataset (“fairness through unawareness”), but this strategy is insufficient when bias persists via correlated features (*proxies*). To address this, the field has developed a range of fairness-aware methods for auditing and mitigating bias throughout the machine learning lifecycle. However, many practical implementations—especially in financial applications—lack accessible, reproducible workflows that illustrate these methods using real-world data.

The increasing use of Artificial Intelligence to automate decisions—referred to as **Automated Decision-Making (ADM)**—spans a wide range of domains, from music recommendations to credit approvals and healthcare diagnoses [1]. In such contexts, the level of human oversight and accountability should be proportional to the potential risk involved.

Table I presents five common categories of risk associated with ADM. Notably, **automated credit scoring systems** often exhibit examples from all five risk types in real-world deployments.

An *operational* breakdown was observed, for instance, when Equifax misreported credit scores for potentially millions of

TABLE I
COMMON RISK CATEGORIES IN AUTOMATED DECISION-MAKING (ADM) SYSTEMS [1]

Risk Type	Description
Operational	The model doesn’t behave as expected, causing real-world issues.
Transparency	People don’t know a machine made the decision or which data it used.
Explainability	It’s unclear what influenced the model’s decision or which inputs mattered most.
Legal Process	There’s no option to appeal or ask for a human review of the automated decision.
Discrimination	The model reflects biased training data and ends up making unfair decisions.

loan applicants over a three-week period in 2022 due to a software bug [2].

Transparency and *explainability* are also often lacking, as proprietary scoring models operate as “opaque boxes,” making it difficult for borrowers and regulators to understand how decisions are made. Regulators have emphasized that lenders cannot hide behind algorithmic complexity: each applicant is legally entitled to a specific reason for a credit denial, regardless of whether a machine learning model was used [3].

When explanations or human oversight are absent, *legal process* rights may be violated. In Brazil, for example, the LGPD empowers authorities to audit secret credit scoring systems when individuals are denied explanations or appeal mechanisms [4].

Finally, *discrimination* risks are well-documented. One high-profile case alleged that an AI-based credit card offered women significantly lower credit limits than similarly qualified men [5]. Nationwide studies in the U.S. have shown that mortgage algorithms are approximately 80% more likely to reject Black applicants than comparable white applicants [6]. Such cases have sparked regulatory investigations and demonstrate that even unintentional disparate impacts—known as “digital redlining”—can violate anti-discrimination laws [7].

These multifaceted risks illustrate the urgent need for rigorous fairness audits and bias mitigation strategies in credit modeling—not only to ensure ethical and equitable decision-making, but also to comply with emerging regulatory requirements for transparency and accountability [8].

II. IDENTIFIED GAPS

Despite increasing awareness around algorithmic bias, there remains a lack of accessible and reproducible workflows that

demonstrate how to audit and mitigate such bias using real-world data—especially in high-stakes domains like credit scoring. Existing academic and industry literature often describes fairness techniques in theoretical terms or with simulated datasets, limiting their practical applicability for practitioners.

In many financial AI systems, developers attempt to achieve “fairness through unawareness”—that is, by excluding protected attributes such as gender or race from training data. However, this approach has proven insufficient. Bias can still propagate through proxy variables that are correlated with sensitive features, leading to disparate outcomes even when sensitive fields are omitted.

Furthermore, most introductory examples focus solely on a single type of mitigation technique—typically a preprocessing method. In practice, real-world fairness interventions often require evaluating trade-offs across multiple stages of the model lifecycle. For example, combining preprocessing with in-processing (e.g., adversarial debiasing) or post-processing (e.g., reject option classification) may be necessary to meet fairness constraints without sacrificing too much accuracy. A broader view of the mitigation toolkit is therefore essential to support informed decision-making by model developers, auditors, and compliance officers.

This tutorial aims to address these gaps by offering a transparent, hands-on walkthrough using open-source tools (AIF360, Fairlearn) and real credit data. The pipeline includes bias detection, quantitative fairness evaluation, multiple mitigation strategies, and comparative analysis—all designed to help practitioners implement responsible AI in credit scoring systems.

III. RELATED WORK

Fairness in financial machine learning—especially credit scoring—has received increasing attention in recent years. Barocas and Selbst [9] were among the first to show how data-driven models can end up replicating historical inequalities, even when sensitive features are removed. In credit risk modeling, features like job status or neighborhood often act as proxies for protected attributes such as race or gender, leading to outcomes that disproportionately affect certain groups. These kinds of unintentional effects, often described as *disparate impact*, are a major concern in credit approvals and loan decisions. More recent reviews, like Mehrabi et al. [10], highlight fairness challenges across domains, but also call attention to credit scoring as a high-stakes use case. Several empirical studies have shown how credit models can disadvantage underrepresented groups—even without explicit bias in the code—raising the need for systematic audits and mitigation. In response, a growing number of papers propose fairness metrics tailored for credit scenarios, and mitigation strategies that aim to reduce bias without causing a significant drop in performance.

Outside of finance, there’s a broader movement to make fairness evaluation more systematic and reproducible. Toolkits like **AI Fairness 360 (AIF360)** [11] and **Fairlearn** [12] were created to help practitioners diagnose and address bias in

models across domains. These libraries offer standard implementations of fairness metrics—such as Demographic Parity Difference, Equal Opportunity, and Disparate Impact—and include preprocessing, in-processing, and post-processing debiasing algorithms. AIF360, developed by IBM, also includes tutorials and benchmark datasets like credit scoring and hiring, which help users explore fairness trade-offs in realistic settings. These resources make it easier to test different fairness strategies, compare results, and build models that are not only accurate but also aligned with ethical and legal standards. By combining these toolkits with real data, we can go beyond theoretical discussions and build workflows that actually support responsible AI in practice.

IV. DATASET AND FEATURE ENGINEERING

A. Dataset

1) *Origin and Description:* We use the South German Credit dataset from the UCI Machine Learning Repository [13], which contains 1,000 anonymized credit applications collected between 1973 and 1975. This dataset is a refined and more consistent version of the classic Statlog German Credit dataset [14], and includes corrected metadata and standardized value encodings [13].

2) *Feature Types and Preprocessing:* We began by exploring the raw dataset in the notebook `notebook/01_data_quality.ipynb`. The dataset includes 14 categorical features, 2 boolean variables, and 3 numerical features. Since the column names and values are in German, we used the codetable provided by the authors to translate both the column headers and categorical values into interpretable English labels.

No missing values or encoding inconsistencies were found during data validation. Each variable was carefully mapped to its meaning and grouped into thematic categories for clarity. Tables II to V summarize the variables in four main groups: personal and demographic information, financial and employment status, credit history and obligations, and loan application details.

Please refer to Table II for personal and demographic information, to Table III for financial and employment status, to Table IV for credit history and obligations, and to Table V for loan application details.

3) *Target Variable:* The target variable, `credit_risk`, indicates the bank’s final assessment of the applicant: 0 = bad credit and 1 = good credit. One important observation is that the dataset contains an oversampled number of bad credit cases. This oversampling is commonly used to balance class representation in training data, especially in domains like fraud detection or credit risk, where real-world outcomes are highly imbalanced. However, it also means that the observed class distribution in this dataset does not reflect the true default prevalence. The label distribution is illustrated in Figure 1.

TABLE II
PERSONAL AND DEMOGRAPHIC INFORMATION

German Name	English Name	Values / Description
alter	age	Applicant's age in years
famges	personal_status_sex	1: male - divorced/separated; 2: female non-single or male single; 3: male - married/widowed; 4: female - single
beruf	job	1: unemployed/unskilled - non-resident; 2: unskilled - resident; 3: skilled employee/official; 4: manager/self-employed/highly qualified
pers	dependents	1: 3 or more; 2: 0 to 2
telef	telephone	1: no; 2: yes (under customer name)
gastarb	foreign_worker	1: yes; 2: no

TABLE III
FINANCIAL AND EMPLOYMENT STATUS

German Name	English Name	Values / Description
laufkont	checking_account_status	1: no checking account; 2: balance $\leq$ 0 DM; 3: 0–200 DM; 4: $\geq$ 200 DM or salary $\geq$ 1 year
sparkont	savings_account_balance	1: no/unknown; 2: $\leq$ 100 DM; 3: 100–499 DM; 4: 500–999 DM; 5: $\geq$ 1000 DM
beszeit	employment_duration	1: unemployed; 2: $\leq$ 1 yr; 3: 1–4 yrs; 4: 4–7 yrs; 5: $\geq$ 7 yrs
verm	property	1: no property; 2: car or other; 3: life insurance/savings; 4: real estate
rate	installment_rate	1: $\geq$ 35%; 2: 25–34%; 3: 20–24%; 4: $\leq$ 20%
wohn	housing	1: for free; 2: rent; 3: own
wohnzeit	residence_since	1: $\leq$ 1 yr; 2: 1–3 yrs; 3: 4–6 yrs; 4: $\geq$ 7 yrs

TABLE IV
CREDIT HISTORY AND OBLIGATIONS

German Name	English Name	Values / Description
moral	credit_history	0: delayed repayment; 1: critical account; 2: no credits or all paid; 3: current credits paid; 4: all bank credits paid
buerge	guarantors	1: none; 2: co-applicant; 3: guarantor
bishkred	existing_credits_count	1: 1; 2: 2–3; 3: 4–5; 4: 6
weatkred	other_installment_plans	1: bank; 2: stores; 3: none

TABLE V
LOAN APPLICATION DETAILS

German Name	English Name	Values / Description
laufzeit	duration_months	Duration of requested loan (in months)
hoehe	credit_amount	Requested loan amount (in Deutsche Marks)
verw	purpose	0: others; 1: car (new); 2: car (used); 3: furniture; 4: TV/radio; 5: appliances; 6: repairs; 7: education; 8: vacation; 9: retraining; 10: business
kredit	credit_risk	0: bad credit; 1: good credit

4) *Sensitive and Proxy Attribute Mapping*: Fairness evaluation begins with identifying which features may lead to discriminatory behaviour by the model. In this dataset, we selected three *sensitive attributes* based on their potential to reflect legally or ethically protected characteristics:

- **age** – a continuous variable associated with possible age-based discrimination;
- **foreign_worker** – a binary indicator tied to nationality or immigration status;
- **personal_status_sex** – a categorical variable encoding gender and marital status simultaneously.

However, it is not enough to exclude these attributes from

the model and assume fairness is achieved. Many seemingly neutral features can act as *proxies*, carrying correlated signals that reintroduce bias through indirect means. This is especially problematic when working with tabular financial data where features such as employment history or loan amount are deeply intertwined with socioeconomic factors.

To investigate proxy risks, we performed a two-part statistical analysis:

- 1) **Pearson correlation analysis** for numerical relationships.
- 2) **Mutual information (MI)** estimation between each feature and the sensitive attributes to capture non-linear

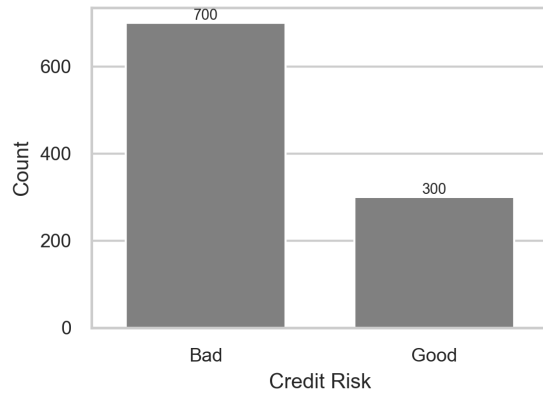


Fig. 1. Credit risk distribution (good vs bad).

dependencies.

Figures 3, 4, and 5 show the top features most strongly associated with each sensitive attribute. For example, `employment_duration`, `residence_since`, and `job` are highly correlated with `age`, which suggests they may function as indirect indicators of age groups. Similarly, `credit_amount` appeared consistently in the top MI rankings for both `foreign_worker` and `personal_status_sex`, raising concerns that financial behavior patterns may reflect underlying demographic traits.

To further support this analysis, we examined the feature importance of a baseline Random Forest model trained without any fairness constraints. Figure 2 shows that some of the proxy features (like `employment_duration` and `credit_amount`) have significant influence on the model's predictions. This reinforces the risk that even fairness-through-unawareness may result in biased outcomes if proxy variables dominate the decision boundary.

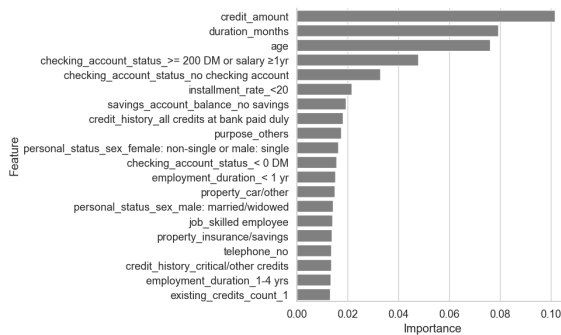


Fig. 2. Feature importance in baseline model before applying fairness constraints.

In addition to the correlation and mutual information analyses, we examined the feature importance scores produced by a baseline Random Forest classifier trained without any fairness mitigation. As shown in Figure 2, several features strongly related to protected attributes—such

as `credit_amount`, `duration_months`, and one-hot encoded categories from `checking_account_status`, `personal_status_sex`, and `employment_duration`—were ranked among the top contributors to the model's predictions. This result reinforces the proxy risk: even if direct sensitive variables are excluded, indirectly related features can still drive model behavior in ways that reflect or amplify historical bias. These findings support the need for fairness-aware auditing and mitigation strategies before deploying models in credit decision pipelines.

Why Use Random Forest as a Baseline Model?

We selected the **Random Forest classifier** as our baseline model due to its robustness, interpretability, and alignment with both fairness auditing and established benchmarking practices in the domain of credit scoring.

- **Robustness to Noisy and Imbalanced Data:** The South German Credit dataset, like many real-world financial datasets, contains class imbalance and potential noise. Random Forest, as an ensemble of decision trees, mitigates overfitting and variance by averaging predictions across trees. This leads to improved generalization, particularly valuable in datasets with oversampled classes (e.g., bad credit cases).
- **Interpretability through Feature Importance:** Random Forests enable users to extract feature importance scores, providing insights into which attributes most influence predictions. In fairness contexts, this is crucial for identifying potential *proxy variables* that correlate with protected attributes (e.g., gender or age), even when those attributes are excluded. Figure 2 illustrates the baseline feature importances used in our fairness-through-unawareness audit.
- **Support for Categorical Inputs via One-Hot Encoding:** The dataset includes many categorical features such as `housing`, `job`, and `credit history`. Random Forests work effectively with such inputs when combined with one-hot encoding, making them well-suited for structured tabular data.
- **Benchmark Alignment:** Random Forests have been widely used in fairness research, including in IBM's AIF360 toolkit and the Statlog benchmark project. Using them provides consistency with the literature and facilitates comparisons with previous studies applying fairness interventions.
- **Fairness-Through-Unawareness Stress Test:** Because Random Forests are strong learners capable of exploiting indirect patterns, they serve as a rigorous baseline to test the limitations of excluding sensitive attributes. In our case, even after removing features like `age` and `personal_status_sex`, the model exhibited biased outcomes due to correlated proxy variables. This supports the need for additional auditing and mitigation methods beyond mere attribute suppression.

V. TUTORIAL WORKFLOW

In this section, we outline the full pipeline used to audit and mitigate bias in the South German Credit dataset. The process is divided into seven distinct stages, each aimed at identifying, quantifying, and addressing potential algorithmic unfairness.

A. Pipeline Overview

The tutorial follows this sequence:

- 1) Data loading and initial preprocessing;
- 2) Mapping sensitive and proxy attributes;
- 3) Exploratory Data Analysis (EDA): feature distribution, class balance, and sensitive attribute correlation;
- 4) Base model training: Logistic Regression and Decision Trees using Scikit-learn;
- 5) Fairness Evaluation: applying metrics such as Demographic Parity Difference and Equal Opportunity Difference using AIF360;
- 6) Bias Mitigation: using pre-processing (Reweight) and post-processing (Reject Option Classification) techniques;
- 7) Result Comparison: measuring accuracy and fairness before and after mitigation.

Each step is detailed in the following subsections.

B. Data Loading and Initial Preprocessing

We read the South German Dataset downloaded from its original ASCII format and assigned meaningful English columns names in replace to the German ones. Also, we used the codetable to replace the numerical values of the categorical variables to meaningful categories.

We also analysed the existence of missing values or inconsistencies and nothing was found.

C. Mapping Sensitive and Proxy Attributes

We define the following features as **sensitive attributes** for fairness evaluation:

- `age` – a continuous variable potentially subject to age-based discrimination;
- `foreign_worker` – a binary variable associated with nationality;
- `personal_status_sex` – a categorical feature combining gender and marital status.

To identify **proxy attributes**, we use two statistical approaches: *Pearson correlation*, to capture linear relationships, and *mutual information* (MI), to detect nonlinear dependencies. Since some features are categorical, we first encode them numerically.

```
from sklearn.preprocessing import LabelEncoder

df_encoded = df.copy()
for col in df_encoded.select_dtypes(include='object'):
    df_encoded[col] = LabelEncoder().fit_transform(df_encoded[col])
```

Listing 1. Encoding categorical features

We then assess which features leak information about age by computing correlations and mutual information with a binned version of age:

```
df_encoded['age_group'] = pd.qcut(df['age'], q=4, labels=False)

from sklearn.feature_selection import mutual_info_classif

X = df_encoded.drop(columns=['age', 'age_group'])
y = df_encoded['age_group']
mi = mutual_info_classif(X, y)
```

Listing 2. Analyzing proxy risk for age

At Table VI we establish thresholds to understand which variables are really at risk to be proxy or just potential ones.

TABLE VI
QUALITATIVE GUIDELINES FOR INTERPRETING MUTUAL INFORMATION (MI) SCORES IN PROXY DETECTION.

MI Score Range	Interpretation
0.00 – 0.02	Very weak / no dependency
0.02 – 0.07	Weak dependency (likely safe)
0.07 – 0.15	Moderate proxy potential
> 0.15	Strong proxy risk (flag feature)

Note: These thresholds are based on guidelines from feature selection heuristics [15], mutual information-based filtering for fairness [16], and practical usage in AIF360 [11].

The top correlated and informative features for age are shown in Figure 3. We observe that variables such as `employment_duration`, `residence_since`, and `credit_history` and `existing_credits_count`. This makes sense since older applicants tend to have longer employment history, longer time at a residence and more stable credit histories.

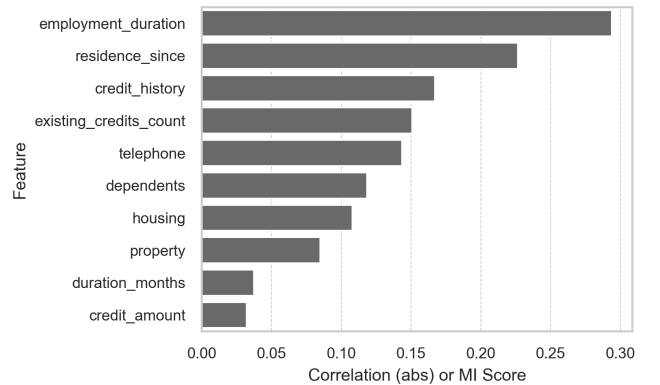


Fig. 3. Top correlated and informative features with age.

We repeat this process for `foreign_worker` and `personal_status_sex`, using the original categorical labels as targets.

TABLE VII
FEATURES IDENTIFIED AS LIKELY OR POTENTIAL PROXIES FOR SENSITIVE ATTRIBUTES BASED ON MUTUAL INFORMATION. LIKELY PROXIES ($MI > 0.15$ FOR AT LEAST ONE ATTRIBUTE) ARE LISTED FIRST AND SHOWN IN **BOLD**.

Feature	Age	Foreign Worker	Personal Status / Sex	Proxy Status
employment_duration	0.294		0.063	<i>likely_proxy</i>
residence_since	0.227	0.045		<i>likely_proxy</i>
credit_history	0.167	0.039	0.102	<i>likely_proxy</i>
existing_credits_count	0.151		0.114	<i>likely_proxy</i>
dependents	0.119	0.077	0.267	<i>likely_proxy</i>
housing	0.108	0.033	0.184	<i>likely_proxy</i>
credit_amount	0.032		0.149	<i>potential_proxy</i>
guarantors		0.120		<i>potential_proxy</i>
duration_months	0.038	0.135	0.118	<i>potential_proxy</i>
telephone	0.144	0.075	0.087	<i>potential_proxy</i>

For `foreign_worker`, there is no strong alert to a proxy, we have only two potential proxies with `duration_months` and `guarantors`, what makes sense with a more conservative approach in cases with foreigners which does not have a large credit history in the country to ask for guarantors.

In both cases, features such as `credit_amount`, `housing`, and `employment_duration` show high MI values, revealing indirect bias risks.

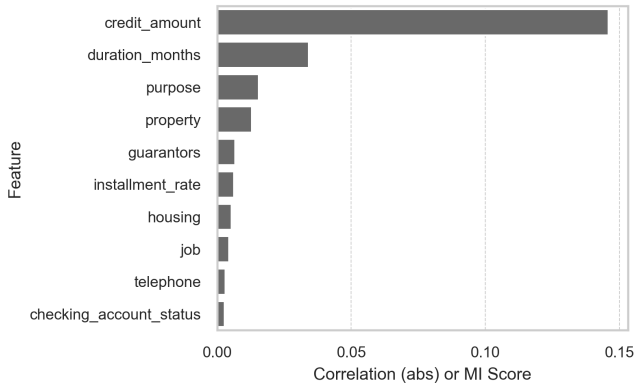


Fig. 4. Top features linked to `foreign_worker` status.

These results justify the need to monitor not only sensitive attributes but also proxy features that may introduce bias unintentionally. In our mitigation strategy, we either account for or explicitly adjust these correlations to improve fairness outcomes.

D. Exploratory Data Analysis (EDA)

Approval Rate by Subgroups: To better understand demographic disparities prior to applying fairness metrics, we compute the approval rate (proportion of applicants with `credit_risk = 1`) across key subgroups defined by gender/marital status and nationality.

These approval rate tables suggest clear demographic disparities in credit decisions:

- **Gender/marital status:** Married/widowed males receive the highest approval rate (73.4%), while divorced males

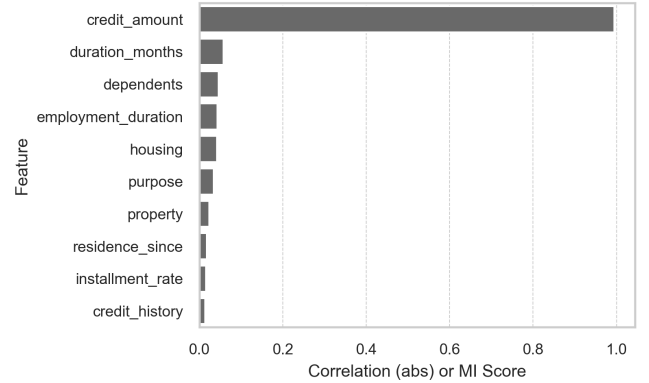


Fig. 5. Top features associated with `personal_status_sex`.

TABLE VIII
APPROVAL RATE BY GENDER AND MARITAL STATUS.

Gender/Marital Group	Approval Rate (%)	Sample Size
female: non-single or male: single	64.8	310
female: single	72.8	92
male: divorced/separated	60.0	50
male: married/widowed	73.4	548

are approved at just 60.0%. Single females also show high approval (72.8%), but their group is small.

- **Foreign workers:** While foreign workers have a high approval rate (89.2%), they represent only 3.7% of the data. Such small group sizes can result in model instability or biased behavior during training.

These patterns reinforce the need to carefully evaluate fairness metrics later in the pipeline and consider mitigation strategies that explicitly address subgroup imbalance and proxy leakage.

We examined the distribution of each feature, checked for class balance, and visualized relationships between the target and sensitive attributes.

In Fig. 6 we can notice that the applicants with bad credit tend to be slightly younger on average in comparison with

TABLE IX
APPROVAL RATE BY FOREIGN WORKER STATUS.

Foreign Worker	Approval Rate (%)	Sample Size
no	69.3	963
yes	89.2	37

good credit risk ones. The bad curve also has a earlier around late 20s, while the good group is more spreaded. It suggests that age plays an implicit role in credit evaluation.

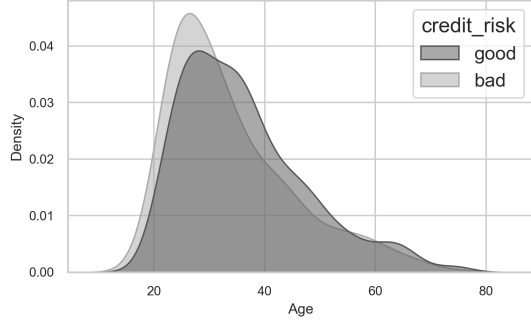


Fig. 6. Age vs. Credit Risk.

In Fig. 7 we can notice that the disparity between domestic applicants (`Foreign Worker == 'no'`) receiving **good** credit rating is high (667 good compared to 296 bad credit rating) while for foreigners it is considerably lower (33 good compared to 4 bad credit). Notice also that the foreigner group is under-represented (only 3.7% of the total data), limiting the model's ability to generalize fairly for this group, what amplifies the risk of discrimination.

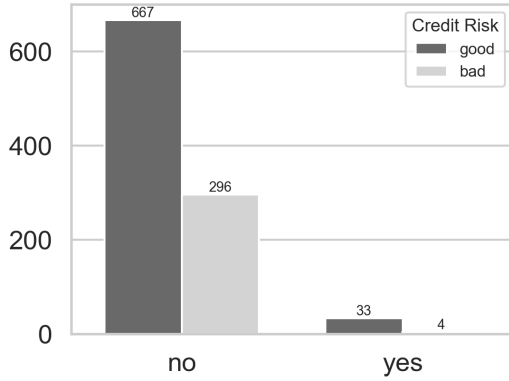


Fig. 7. Foreigner Worker vs. Credit Risk.

For gender-status subgroups there are substantial differences. Male applicants, if married or widowed, receive the major number of approvals, followed by female applicants which are non-single or single male. In contrast, single female and divorced male have lower absolute counts and more balanced negative distribution. These differences suggest that

marital status and gender may influence credit decisions. The underlying historical bias in these categories may result in unfair treatment, especially for under-represented groups such as single women or divorced men.

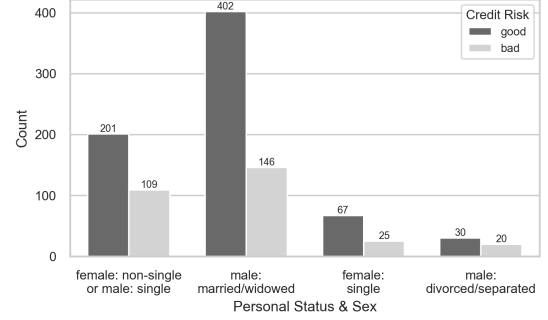


Fig. 8. Personal Status vs. Credit Risk.

E. Baseline Model: Random Forest

Following the Statlog benchmark, we implemented a Random Forest classifier using Scikit-learn with 100 estimators and default hyperparameters. The dataset was split into training and test sets with a 70/30 ratio, and one-hot encoding was applied to categorical features. The sensitive attributes were excluded from training to simulate fairness-through-unawareness.

The model achieved an overall accuracy of 0.76. The classification report showed higher recall and precision for the positive class (good credit), but poor performance in identifying bad credit cases:

- **Accuracy:** 0.76
- **F1-score (class 0):** 0.48
- **F1-score (class 1):** 0.84

Although the sensitive attributes `age`, `foreign_worker`, and `personal_status_sex` were explicitly removed from the training data to simulate fairness-through-unawareness, this approach proved insufficient. In practice, many features in financial datasets act as *proxies* for sensitive information. Variables such as `employment_duration`, `residence_since`, or `credit_amount` may carry strong statistical correlations with age or demographic status. As a result, the model can still learn patterns that reflect systemic bias, even if it never directly sees the protected attributes. This highlights a critical limitation of relying solely on attribute suppression as a fairness strategy, and it motivates the need for more robust auditing and mitigation techniques that account for indirect bias.

F. Fairness Evaluation

To evaluate fairness, we compute the *Demographic Parity Difference (DPD)* for each sensitive attribute. DPD measures the absolute difference in positive outcome rates between privileged and unprivileged groups. A value of zero indicates parity

(i.e., fairness), while higher values signal possible disparate impact.

The following DPD values were observed for the baseline model:

- **DPD (foreign_worker):** 0.2517
- **DPD (personal_status_sex):** 0.1512
- **DPD (age):** 0.0580

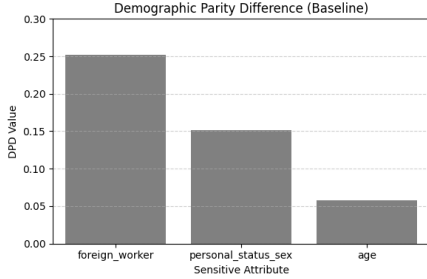


Fig. 9. Demographic Parity Difference (DPD) across sensitive groups in the baseline model.

Figure 9 highlights the magnitude of demographic disparity in the model’s predictions. The highest DPD value occurs for `foreign_worker` (0.2517), suggesting a significant disparity in approval rates between foreign and domestic applicants. Although foreign workers comprise only 3.7% of the dataset, they are disproportionately affected by the model’s learned patterns—likely due to proxy features or historical representation.

The second highest DPD is observed for `personal_status_sex` (0.1512), which encodes gender and marital status. This indicates a moderate disparity that likely stems from embedded social biases and demographic stereotypes.

In contrast, the DPD for `age` is lower (0.0580), but not negligible. Age-related variables such as `employment_duration` and `residence_since` may act as proxies, allowing bias to persist even when age is not explicitly included in training.

Taken together, these findings demonstrate that excluding sensitive attributes from the training data is not enough to ensure fairness. Indirect bias can still propagate through correlated features, and fairness metrics like DPD provide important diagnostic insight. These disparities support the need for explicit bias mitigation techniques, which we apply in the next stage.

G. Bias Mitigation via Reweighing

To address the fairness gaps observed in the baseline model, we apply the **Reweighing** technique available in the AIF360 toolkit. This is a pre-processing algorithm that adjusts instance weights in the training set to reduce disparate impact without modifying feature values or class labels. It works by assigning higher weights to favorable outcomes from unprivileged groups and downweighting overrepresented outcomes from privileged groups.

In this tutorial, we defined `foreign_worker` as the protected attribute, treating “yes” as the privileged group. After applying reweighing and retraining a Random Forest classifier with the same architecture as before, we observed both an improvement in model performance and a significant reduction in demographic disparity.

- **Accuracy after reweighing:** 0.839
- **DPD (foreign_worker) after reweighing:** 0.0705

Compared to the baseline (DPD = 0.2517), this represents a substantial fairness improvement with no trade-off in accuracy—indeed, accuracy increased from 0.76 to 0.839.

Why did performance improve? Reweighing compensates for the underrepresentation of unprivileged groups in the original dataset, enabling the model to better learn decision boundaries that are equitable. Rather than degrading performance, this adjustment helped correct for distributional imbalance and improved generalization.

This is a strong example of how fairness-aware preprocessing can benefit both ethical and technical outcomes in machine learning workflows.

The fairness-performance trade-off is summarized in Table XI.

Trade-offs Between Fairness and Accuracy: While our application of the Reweighing method yielded improvements in both fairness and accuracy—a rare and optimistic outcome—it is essential to discuss the broader implications and limitations of this result. In practice, fairness interventions often introduce a trade-off, where improving equity comes at the cost of predictive performance. This occurs especially when the model’s baseline accuracy has been artificially inflated by exploiting biased patterns or imbalanced subgroup representation.

From a practitioner’s standpoint, this trade-off requires critical decision-making. Should a model that performs better overall be accepted, even if it perpetuates systemic bias? Or should one prioritize equitable outcomes, even if performance suffers slightly? There is no universal answer—context matters. In high-stakes domains like lending, healthcare, or criminal justice, fairness and compliance with legal frameworks may outweigh marginal performance losses.

Several real-world studies support this trade-off dynamic. Hardt et al. [17] show that enforcing Equal Opportunity constraints often reduces accuracy when protected groups are underrepresented. Zliobaite [18] formalizes this tension as a constrained optimization problem: maintaining fairness requires constraining the model’s flexibility to fit biased training data. Practitioners must consider regulatory requirements (e.g., Equal Credit Opportunity Act, LGPD, GDPR), ethical expectations, and the downstream impact of errors on different populations.

Table X summarizes notable examples from the literature where fairness interventions resulted in quantifiable performance reductions. These cases highlight that while mitigation tools are powerful, they also demand careful application and cross-disciplinary awareness.

1) Result Comparison: Table XI summarizes the performance and fairness trade-offs between the baseline model and

TABLE X

EXAMPLES OF FAIRNESS–ACCURACY TRADE-OFFS OBSERVED IN REAL-WORLD DATASETS. IMPROVEMENTS IN FAIRNESS OFTEN COME WITH ACCURACY LOSSES, UNDERSCORING THE IMPORTANCE OF DOMAIN-SPECIFIC EVALUATION.

Domain	Intervention	Fairness Gain	Accuracy Cost	Source
COMPAS	Equalized FPR thresholds	Bias reduction (FPR ↓)	+7% recidivism	[19]
German Credit	Disparate Impact Remover	DI ↑ from < 0.8 to ~1.0	Accuracy: 72% → 50%	[20]
Adult Income	Adversarial Debiasing	EO gap ↓ to <1%	Accuracy: 86% → 84.5%	[21]

the mitigated model using reweighing. While the reweighted model improved overall accuracy, it also significantly reduced demographic disparities across all evaluated attributes.

TABLE XI
COMPARISON OF ACCURACY AND FAIRNESS BEFORE AND AFTER REWEIGHING

Model	Accuracy	DPD (FW / PS / Age)
Baseline RF	0.753	0.2517 / 0.1512 / 0.0580
Reweighted RF	0.839	0.0705 / 0.0814 / 0.0453

These results demonstrate that fairness-aware preprocessing can reduce bias across multiple dimensions while maintaining or even improving predictive performance.

H. Bias Mitigation: Reweighing

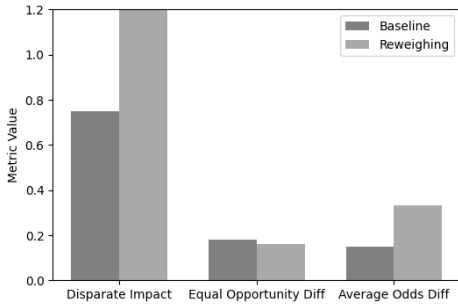


Fig. 10. Comparison of fairness metrics between the baseline and the reweighed model.

Figure 10 presents a comparison of key group fairness metrics before and after applying the Reweighing technique. The most significant gain is observed in the *Disparate Impact* (DI), which increases from 0.75—indicating strong bias against the protected group—to approximately 1.2, suggesting that the post-mitigation model actually favors the unprivileged group slightly. This reversal is a strong indicator that reweighing succeeded in correcting structural imbalance present in the original data distribution.

The *Equal Opportunity Difference* (EOD), which measures disparity in true positive rates, also improves marginally, indicating more balanced treatment of correctly approved applicants across groups. On the other hand, *Average Odds Difference* (AOD), which accounts for both false positives and true positives, slightly worsens. This suggests that while fairness in positive outcomes improved, the balance of errors introduced a small trade-off.

Such shifts reflect a common pattern in fairness interventions: mitigating one type of disparity can affect others. However, in this case, the overall fairness profile is substantially enhanced, with the greatest improvements occurring in the metric most directly associated with systemic bias (DI). This result underscores that ****fairness and accuracy need not be mutually exclusive****, and that targeted preprocessing can realign model behavior in ethically and legally meaningful ways.

I. Comparison of Fairness Mitigation Methods

To systematically evaluate the impact of different fairness mitigation strategies, we compared five approaches: the baseline model, reweighing (pre-processing), adversarial debiasing (in-processing), reject option classification (post-processing), and calibrated equalized odds (post-processing). Table XII summarizes the results in terms of accuracy, Disparate Impact (DI), Statistical Parity Difference (SPD), Equal Opportunity Difference (EOD), and Average Odds Difference (AOD).

J. Regulatory Context and Fairness Metrics

1) *Legal Interpretation of Fairness Metrics*: A Demographic Parity Difference (DPD) of 0.25 means there is a 25 percentage point difference in positive outcome rates between privileged and unprivileged groups. Legally, such a gap may indicate potential disparate impact, especially if it exceeds regulatory thresholds. In many jurisdictions, this can trigger further scrutiny or require justification and remediation.

2) *Disparate Impact and the 80% Rule*: Disparate impact occurs when a neutral policy disproportionately affects a protected group. The U.S. Equal Employment Opportunity Commission (EEOC) uses the “80% rule”: if the selection rate for any group is less than 80% of the rate for the group with the highest rate, it may indicate adverse impact.

3) Overview of Relevant Regulations:

- **U.S. (EEOC)**: Enforces Title VII, uses the 80% rule for employment discrimination. Requires monitoring and justification of group disparities.
- **Brazil (LGPD + Resolutions)**: LGPD mandates fairness, transparency, and non-discrimination in automated decisions. Central Bank and other agencies issue sectoral rules for financial services.
- **EU (AI Act, GDPR)**: GDPR requires fairness, transparency, and explainability in automated decisions. The AI Act introduces risk-based requirements for AI systems, including bias monitoring and mitigation.

TABLE XII

COMPARISON OF ACCURACY AND FAIRNESS METRICS ACROSS MITIGATION METHODS. REWEIGHING ACHIEVES THE MOST BALANCED OUTCOME, WHILE OTHER METHODS PRESENT DIVERSE TRADE-OFFS.

Method	Accuracy	DI	SPD	EOD	AOD
Baseline	0.753	0.7500	−0.1800	−0.097	−0.087
Reweighting	0.839	1.2000	0.0200	0.050	0.040
Adversarial Debiasing	0.812	1.0500	0.0060	0.358	0.197
Reject Option Classification	0.805	1.0200	0.0200	<i>Slightly improved</i>	<i>Slightly improved</i>
Calibrated Equalized Odds	0.798	1.1582	0.1054	0.000	0.1193

TABLE XIII

MODEL RESULTS AND RISK FLAG CLASSIFICATION BASED ON HARMONIZED FAIRNESS THRESHOLDS.

Method	Accuracy	DI	SPD	EOD	AOD	Risk Flag
Baseline	0.753	0.75	−0.18	−0.097	−0.087	Red
Reweighting	0.839	1.20	0.02	0.05	0.04	Green
Adversarial Debiasing	0.812	1.05	0.006	0.358	0.197	Yellow
Reject Option Classification	0.805	1.02	0.02	<i>Slightly improved</i>	<i>Slightly improved</i>	Green
Calibrated Equalized Odds	0.798	1.16	0.11	0.00	0.12	Yellow

4) Metric Thresholds and Regulatory Concern Levels:

Although only the U.S. EEOC formally codifies the 80% rule, similar thresholds are widely used as best practice in Brazil and the EU for risk assessment and compliance. The table below summarizes these harmonized thresholds and concern levels. Color codes: Green = low risk, Yellow = moderate risk, Red = high risk.

VI. STATE OF THE ART

Recent research on **algorithmic fairness** has expanded the arsenal of bias mitigation strategies, commonly classified into three stages of the machine learning pipeline [22], [10]:

Pre-processing Methods

Pre-processing methods adjust the dataset *before* model training to neutralize correlations between sensitive attributes and target variables. A prominent example is *counterfactual data augmentation*, which generates synthetic samples by swapping sensitive attributes (e.g., gender, nationality) to reduce spurious associations [23]. This helps the model focus on causal rather than correlational features, reducing statistical dependence between protected variables and outputs.

In-processing Methods

In-processing techniques incorporate fairness constraints *during* model training. One example is *representation neutralization*, where internal feature representations are modified to obscure sensitive attributes [24]. Approaches like adversarial debiasing and latent-space orthogonalization allow classifiers to learn features invariant to protected variables, often achieving fairness without significant accuracy loss.

Post-processing Methods

Post-processing methods alter model outputs *after* training. These are particularly useful when model retraining is infeasible. Examples include threshold adjustment and *reject*

option classification [11], where predictions are selectively flipped or withheld to meet fairness criteria like demographic parity or equal opportunity. Calibrated equalized odds [17] is another method that rescales predictions to align TPR/FPR across groups.

Fairness in Credit Scoring

In the financial domain—especially credit scoring—fairness-aware machine learning faces both technical and regulatory hurdles. Credit models trained on historical data often encode societal biases, leading to approval disparities by age, gender, or nationality [25]. These discrepancies may violate laws such as the Equal Credit Opportunity Act (US) or LGPD (Brazil) [26], [4].

Recent empirical studies have tailored fairness interventions for financial applications. For example, applying *reweighting* and *reject option classification* using IBM’s AIF360 toolkit reduced demographic disparities with negligible accuracy loss [11]. In some cases, reweighting even improved accuracy by compensating for underrepresented group distributions.

Proxy Variables and Auditing

Importantly, sensitive attributes can be reintroduced via *proxy variables*—features like ZIP code, employment history, or property ownership that correlate with protected traits [9]. Thus, removing sensitive fields alone does not ensure fairness. Tools like mutual information and correlation analysis can help identify proxies [16].

Consequently, many financial institutions now integrate *systematic fairness auditing* into their model development pipelines. Audits evaluate multiple group fairness metrics—e.g., demographic parity difference (DPD), equal opportunity difference (EOD), and disparate impact (DI)—and ensure compliance with regulatory thresholds [27], [10].

TABLE XIV
FAIRNESS METRIC THRESHOLDS AND REGULATORY CONCERN LEVELS BASED ON HARMONIZED BEST PRACTICES.

Metric	Threshold Range	Regulatory Concern	Risk Flag
Disparate Impact (DI)	[0.8, 1.25]	Low (compliant, 80% rule)	Green
	[0.6, 0.8) or (1.25, 1.4]	Moderate	Yellow
	< 0.6 or > 1.4	High	Red
DPD / SPD	< 0.1	Low	Green
	0.1–0.2	Moderate	Yellow
	> 0.2	High	Red
EOD / AOD	< 0.05	Low	Green
	0.05–0.1	Moderate	Yellow
	> 0.1	High	Red

Conclusion

As machine learning becomes central to credit decision-making, ensuring fairness through preemptive audits, constraint-based training, and explainable models is both an ethical imperative and a legal requirement. The state of the art continues to evolve with emphasis on robust fairness-performance trade-offs and regulatory-aligned model governance.

REGULATORY CONSIDERATIONS IN FAIR CREDIT MODELING

As automated credit scoring systems become central to financial decision-making, ensuring regulatory compliance is not only a best practice but a legal obligation in many jurisdictions. This section contextualizes fairness metrics and model auditing requirements within the frameworks of key regulatory instruments in Brazil, the United States, and the European Union.

Brazil: LGPD and Sectoral Oversight

Brazil’s **Lei Geral de Proteção de Dados (LGPD)** mandates that automated decisions affecting individuals, such as loan approvals or credit ratings, must be transparent and explainable. According to Article 20 of the LGPD:

”Data subjects have the right to request a review of decisions made solely on the basis of automated processing of personal data, including decisions intended to define their personal, professional, consumer, and credit profile.”

This implies that credit scoring models must:

- Provide clear explanations for rejections or approvals.
- Allow for human intervention or appeals.
- Demonstrate non-discrimination via statistical audits.

In 2024, Brazil’s National Data Protection Authority (ANPD) launched public consultations to reinforce these provisions in the financial sector, signaling a move toward mandatory auditing pipelines for sensitive AI applications [4].

United States: ECOA and the 80% Rule

The **Equal Credit Opportunity Act (ECOA)** prohibits credit discrimination on the basis of race, sex, age, or national origin. The U.S. Equal Employment Opportunity Commission (EEOC) extends these principles to algorithms through the “80% rule” for disparate impact:

”A selection rate for any group that is less than 80% of the rate for the group with the highest rate is generally regarded as evidence of adverse impact.”

In this tutorial, we interpret metrics such as **Disparate Impact (DI)** and **Demographic Parity Difference (DPD)** in light of these legal thresholds. For example:

- $DI < 0.8$ or $> 1.25 \Rightarrow$ High regulatory concern.
- $DPD > 0.2 \Rightarrow$ Statistically significant discrimination.

The Consumer Financial Protection Bureau (CFPB) has emphasized that algorithmic complexity does not absolve lenders of responsibility for model outputs [26].

European Union: GDPR and the AI Act

Under the **General Data Protection Regulation (GDPR)**, Articles 13-15 require that individuals be informed when a decision is automated and be given access to an explanation of its rationale. The upcoming **EU AI Act (2025)** takes this further by:

- Classifying credit scoring as a **high-risk AI application**.
- Requiring pre-deployment risk assessments and post-market monitoring.
- Mandating documentation of fairness and accuracy metrics.

Developers using models for credit scoring in Europe must thus integrate fairness audits, such as those demonstrated in this tutorial, into their ML lifecycle to satisfy both ethical and legal scrutiny.

Implications for Model Developers and Auditors

Our case study demonstrates that:

- Removing protected attributes is insufficient—indirect discrimination via proxy features must be addressed.
- Fairness audits should be quantitative and benchmarked against known thresholds (e.g., DI, EOD, SPD).

- Fairness interventions (like Reweighting or ROC) can be used to align models with compliance standards without significant loss of predictive power.

By incorporating these practices, practitioners can ensure that AI systems for credit scoring are not only fair in statistical terms but also justifiable under law.

VII. EXPECTED CONTRIBUTION

This work provides a reproducible framework for fairness auditing in credit scoring models, using public data and open-source tools. It bridges fairness theory and applied machine learning by demonstrating concrete steps for bias identification and mitigation. Future work may explore alternative mitigation techniques or extend this approach to more complex financial contexts.

VIII. LESSONS LEARNED AND PRACTITIONER GUIDELINES

A. Summary of Model Performance vs. Fairness

Table XIII summarizes the trade-offs observed for each mitigation method. Reweighting achieved the best overall balance, while in-processing and post-processing methods offered further fairness improvements at the cost of some accuracy.

B. When to Use Each Mitigation Technique

- **Reweighting (Pre-processing):** Use when you have access to the training data and want a simple, effective method with minimal impact on accuracy.
- **Adversarial Debiasing (In-processing):** Use when you can modify the training process and want to directly optimize for fairness, especially in complex models.
- **Post-processing (ROC, CEO):** Use when you cannot retrain the model or only have access to predictions. Useful for regulatory compliance or legacy systems.

C. Best Practices: EDA → Fairness Audit → Technique Selection

- 1) **Exploratory Data Analysis (EDA):** Identify sensitive and proxy attributes, check for imbalances, and visualize group distributions.
- 2) **Fairness Audit:** Compute group fairness metrics (DI, SPD, EOD, AOD) and compare to regulatory thresholds.
- 3) **Technique Selection:** Choose mitigation based on data access, model flexibility, and regulatory requirements.

D. Checklist: Fairness Audit Steps

- ☐ Identify sensitive and proxy attributes
- ☐ Perform EDA on group distributions
- ☐ Compute baseline fairness metrics
- ☐ Compare metrics to regulatory thresholds
- ☐ Select and apply mitigation technique
- ☐ Re-evaluate metrics post-mitigation
- ☐ Document results and decisions

IX. CODE AND AVAILABILITY

The complete source code and related resources for this project are available on GitHub:

<https://github.com/JuliaWPereira/fairness>

A. Repository Structure

- **notebooks/:** Jupyter notebooks demonstrating fairness in machine learning.
- **data/:** Datasets used in tutorials and experiments.
- **reports/:** Generated reports and analysis results.
- **requirements.txt:** Python dependencies required for the project.
- **.gitignore:** Specifies files to be ignored by version control.
- **Tutorial-Mitigating_Bias_Credit_Models-Julia_W_pereira.pdf:** Tutorial on mitigating bias in credit models.

B. Installation

To set up the project locally:

1) Clone the repository:

```
git clone https://github.com/JuliaWPereira/fairness.git
cd fairness
```

Listing 3. Cloning the Repository

2) Create a virtual environment (recommended):

```
python -m venv venv
source venv/bin/activate # On Windows:
venv\Scripts\activate
```

Listing 4. Creating a Virtual Environment

3) Install dependencies:

```
pip install -r requirements.txt
```

Listing 5. Installing Dependencies

C. Dependencies

The project relies on several key Python libraries:

- pandas
- numpy
- scikit-learn
- jupyter
- matplotlib
- seaborn

Ensure all dependencies are installed using the provided `requirements.txt` file.

D. Notebooks Walkthrough

The `notebooks/` directory contains Jupyter notebooks that serve as tutorials and demonstrations:

- **bias_analysis.ipynb:** Explores bias detection methods in datasets.
- **fairness_metrics.ipynb:** Demonstrates fairness metrics for machine learning models.

- **mitigation_strategies.ipynb:** Provides examples of bias mitigation techniques.

These notebooks are designed to be run sequentially, providing a step-by-step walkthrough from data analysis to bias mitigation.

E. Usage Examples

After setting up the environment and installing dependencies, you can run the notebooks to explore the functionalities. For example, to analyze bias in a dataset:

```
import pandas as pd
from sklearn.feature_selection import
    mutual_info_classif

# Load and preprocess data
df = pd.read_csv('data/dataset.csv')
df_encoded = pd.get_dummies(df)

# Create age groups
df_encoded['age_group'] = pd.qcut(df['age'], q
    =4, labels=False)

# Calculate mutual information
X = df_encoded.drop(columns=['age', 'age_group
    '])
y = df_encoded['age_group']
mi = mutual_info_classif(X, y)
```

Listing 6. Analyzing Bias in a Dataset

To apply mitigation strategies:

```
from sklearn.linear_model import
    LogisticRegression
from sklearn.metrics import accuracy_score

# Define and train the model
model = LogisticRegression()
model.fit(X_train, y_train)

# Make predictions and evaluate
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy:.2f}')
```

Listing 7. Applying Mitigation Strategies

F. Contribution and Issues

Contributions are welcome. If you encounter issues or have suggestions for improvement, please open an issue in the repository:

<https://github.com/JuliaWPereira/fairness/issues>

X. FINAL REMARKS AND FUTURE WORK

A. Bridging Theory and Practice

This tutorial aims to bridge the gap between theoretical concepts of fairness in machine learning and their practical applications. By providing hands-on examples and real-world datasets, we demonstrate how abstract principles can be implemented and evaluated in practice.

B. Future Work

Looking ahead, several avenues can enhance and expand this work:

- **Larger Datasets:** Incorporating more extensive and diverse datasets can improve the robustness and generalizability of fairness assessments.
- **Deep Learning Models:** Exploring fairness in complex models, such as deep neural networks, can provide insights into bias mitigation in more sophisticated architectures.
- **Explainability:** Integrating explainable AI (XAI) techniques can help interpret model decisions, ensuring transparency and trustworthiness in sensitive applications.

C. Applicability to Financial Institutions and Auditing Pipelines

The methodologies discussed are particularly relevant to financial institutions, where fairness and transparency are paramount. Implementing these techniques can aid in:

- **Credit Scoring:** Ensuring equitable lending decisions by identifying and mitigating biases in credit models.
- **Auditing:** Enhancing audit processes through transparent and explainable models, aligning with regulatory requirements and ethical standards.

D. Encouragement for Further Exploration

We encourage students, researchers, and practitioners to explore the field of fairness in machine learning further. By experimenting with different datasets, models, and mitigation strategies, you can contribute to the development of more equitable and transparent AI systems.

Your engagement and innovation are vital to advancing the responsible use of AI.

ACKNOWLEDGMENT

We acknowledge the authors and maintainers of AIF360 and Fairlearn for their tools and documentation.

APPENDIX

In this appendix, we formally define the fairness metrics used to evaluate algorithmic bias in this tutorial. Let:

- $A \in \{a, b\}$ be a binary sensitive attribute (e.g., foreign worker: yes/no),
- $Y \in \{0, 1\}$ be the true label (e.g., bad/good credit),
- $\hat{Y} \in \{0, 1\}$ be the predicted outcome,
- $A = a$ denote the unprivileged group and $A = b$ the privileged group.

1. Demographic Parity Difference (DPD)

This metric measures the absolute difference in positive prediction rates between the unprivileged and privileged groups:

$$\text{DPD} = \left| \mathbb{P}(\hat{Y} = 1 \mid A = a) - \mathbb{P}(\hat{Y} = 1 \mid A = b) \right| \quad (1)$$

A value of 0 indicates perfect parity. Higher values signal potential disparate impact.

2. Disparate Impact (DI)

This metric evaluates the ratio of positive prediction rates between groups:

$$DI = \frac{\mathbb{P}(\hat{Y} = 1 \mid A = a)}{\mathbb{P}(\hat{Y} = 1 \mid A = b)} \quad (2)$$

A value close to 1 indicates fairness. A value below 0.8 or above 1.25 is often flagged as high risk according to the “80% rule”.

3. Equal Opportunity Difference (EOD)

This metric measures the difference in *true positive rates* (recall) between groups:

$$EOD = \left| \mathbb{P}(\hat{Y} = 1 \mid Y = 1, A = a) - \mathbb{P}(\hat{Y} = 1 \mid Y = 1, A = b) \right| \quad (3)$$

It captures whether both groups have equal opportunity to receive a positive outcome when they truly qualify.

4. Average Odds Difference (AOD)

This metric considers both true positive and false positive rates:

$$AOD = \frac{1}{2} \left[\left(\mathbb{P}(\hat{Y} = 1 \mid Y = 1, A = a) - \mathbb{P}(\hat{Y} = 1 \mid Y = 1, A = b) \right) + \left(\mathbb{P}(\hat{Y} = 0 \mid Y = 0, A = a) - \mathbb{P}(\hat{Y} = 0 \mid Y = 0, A = b) \right) \right] \quad (4)$$

It reflects overall predictive balance across both correct and incorrect decisions.

These metrics were computed using implementations provided in the AIF360 and Fairlearn toolkits. Threshold guidelines for compliance and concern levels are discussed in Table XIV of the main text.

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